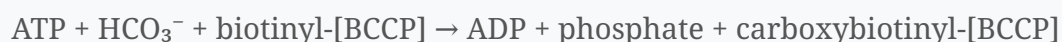


Functional Annotation of *accC* (Biotin Carboxylase, Q88QD6) in *Pseudomonas putida* KT2440

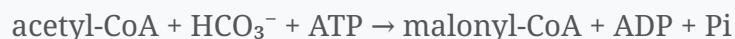
1. Summary / Answer to the Research Question

accC (gene locus PP_0558; UniProt Q88QD6) of *Pseudomonas putida* KT2440 encodes **biotin carboxylase (BC)**, the **biotin carboxylase subunit A of acetyl-CoA carboxylase (ACC)** (EC 6.3.4.14). Its primary, and essentially sole, molecular function is to catalyze the **ATP-dependent carboxylation of the biotin prosthetic group** that is covalently attached to the biotin carboxyl carrier protein (BCCP/AccB), using **bicarbonate (HCO_3^-)** as the carboxyl donor and requiring **Mg^{2+}** . This is the **first of the two half-reactions of acetyl-CoA carboxylase**, which together constitute the **first committed and rate-limiting step of fatty acid (membrane phospholipid) biosynthesis**. BC is a **soluble, cytoplasmic** enzyme that functions as a **homodimer** within the transient multi-subunit ACC complex.

Reaction catalyzed (BC half-reaction):



Overall ACC pathway (UniPathway UPA00655, "malonyl-CoA from acetyl-CoA: step 1/1"):



2. Gene/Protein Identity Verification

The gene symbol ***accC*** is **unambiguous** and universally denotes **biotin carboxylase** across bacteria; there is no conflicting gene using this symbol. The assignment is fully consistent across every line of evidence:

Attribute	Value (UniProt Q88QD6 / this analysis)
Protein	Biotin carboxylase; ACC biotin carboxylase subunit A
EC number	6.3.4.14 (biotin carboxylase)
Gene / locus	<i>accC</i> / PP_0558
Organism	<i>Pseudomonas putida</i> KT2440 (taxid 160488)
Length	451 aa
Key domains	Biotin_carboxylation (res. 4–448); ATP-grasp (res. 123–320); Pfam PF00289, PF02785, PF02786; InterPro IPR004549, IPR011761, IPR013815, IPR005481
Pathway	Lipid metabolism → malonyl-CoA biosynthesis (UPA00655), step 1/1
Ligands (keywords)	ATP, biotin, Mg ²⁺
KEGG / BioCyc	ppu:PP_0558 / PPUT160488:G1G01-608-MONOMER

Because the UniProt protein-existence level is "**Predicted**", the functional annotation here rests primarily on **orthology and conservation** to biochemically and structurally characterized biotin carboxylases (*E. coli*, *H. influenzae*), reinforced below by direct sequence analysis.

3. Primary Function: The Reaction Catalyzed

Acetyl-CoA carboxylase in *P. putida*, as in other Gram-negative bacteria, is a **multi-component (dissociable) enzyme** built from four separate polypeptides: biotin carboxylase (AccC), biotin carboxyl carrier protein (BCCP/AccB), and the two carboxyltransferase subunits (AccA, AccD). Catalysis proceeds in two spatially separated half-reactions [PMID 16707089, 26020841]:

1. **BC half-reaction (this gene, AccC):** ATP-dependent carboxylation of the biotin covalently bound to BCCP, using bicarbonate as the CO₂ source.
2. **Carboxyltransferase half-reaction (AccA/AccD):** transfer of the carboxyl group from carboxybiotin-BCCP to acetyl-CoA, yielding **malonyl-CoA**.

The BC subunit "**catalyzes the ATP-dependent carboxylation of biotin using bicarbonate as the carboxylate source and has a distinct architecture that is characteristic of the ATP-grasp superfamily of enzymes**" [PMID 11346647]. Because ACC "**catalyzes the first committed and regulated step in fatty acid biosynthesis in bacteria**" [PMID 16707089] and is essential for membrane lipid production, it — and BC specifically — is a validated antibacterial drug target [PMID 16707089, 28469974].

Substrate specificity. BC is highly specific: it requires **MgATP** (not other NTPs efficiently), **bicarbonate** (the true substrate; free CO₂ is not used), and the **biotin** moiety (physiologically presented on BCCP; small free-biotin analogues such as biotin or N-1-methylbiotin are used in vitro). Steady-state kinetics of the *E. coli* enzyme show that "**MgATP and bicarbonate add to the enzyme randomly, followed by addition of biotin**," with both MgATP and bicarbonate binding in rapid equilibrium and a catalytic base of $pK \approx 6.6$ in the active site [PMID 2971391].

4. Catalytic Mechanism (ATP-grasp fold, carboxyphosphate intermediate)

BC belongs to the **ATP-grasp superfamily** (which also includes D-Ala-D-Ala ligase, glutathione synthetase, carbamoyl-phosphate synthetase, and N⁵-CAIR synthetase) [PMID 11346647]. Its mechanism proceeds through an **unstable carboxyphosphate intermediate**:

1. **ATP activates bicarbonate.** The γ -phosphate of ATP is transferred to bicarbonate, forming **carboxyphosphate** + ADP. Isotope-tracer and positional-isotope-exchange experiments on the *E. coli* BC subunit showed that the bicarbonate-dependent, biotin-independent ATPase "**goes via carboxyphosphate**," with ¹⁸O transferred from bicarbonate into product Pi [PMID 2976600].
2. **Carboxyphosphate decomposes** to CO₂ + phosphate. The liberated phosphate acts as the general base that deprotonates the N1' of biotin, and the resulting enolate/nucleophilic biotin attacks CO₂. High-resolution *Haemophilus influenzae* BC structures with substrate, intermediate-analogue, and product complexes support "**direct transfer of CO₂ from the carboxyphosphate intermediate to biotin**," with bicarbonate, phosphate, and the carboxyl group all occupying the same tetrahedral-stabilizing pocket [PMID 26020841].
3. Crystal structures of wild-type *E. coli* BC with biotin, bicarbonate and Mg-ADP indicate that **Glu296 is the general base that extracts the proton from bicarbonate and Arg338**

stabilizes the enolate-biotin intermediate; a mobile "B domain" closes over the active site to align the nucleotide for catalysis [PMID 19213731].

Conserved catalytic architecture in *P. putida* (direct evidence from this analysis). A global Needleman–Wunsch alignment of Q88QD6 (451 aa) against the biochemically characterized *E. coli* AccC (P24182, 449 aa) gives **70.5 % identity (316/448)**. Critically, **every experimentally validated catalytic and ATP-binding residue is strictly conserved**:

Function	<i>E. coli</i> residue	<i>P. putida</i> (Q88QD6) residue	Status
ATP binding	Lys116	Lys119	conserved
ATP binding	Lys159	Lys162	conserved
ATP binding	His209	His212	conserved
ATP binding / catalysis	Glu276	Glu279	conserved
Active-site Arg	Arg292	Arg295	conserved
General base (bicarbonate)	Glu296	Glu299	conserved
Enolate-biotin stabilization	Arg338	Arg341	conserved

The intact ATP-grasp fold (UniProt domain res. 123–320) plus this complete, unbroken catalytic constellation constitute strong structural/evolutionary evidence that Q88QD6 is a fully active biotin carboxylase operating by the canonical carboxyphosphate mechanism.

5. Subcellular Localization and Quaternary Structure

- **Localization:** BC is a **soluble cytoplasmic** protein. In Gram-negative bacteria the four ACC components are separate cytosolic polypeptides that assemble into a large but **labile, transient complex**; BC has no signal peptide or membrane anchor (consistent with the intracellular site of fatty-acid initiation). The malonyl-CoA produced is subsequently handed to the cytoplasmic type II fatty-acid synthase (FAS II) machinery.
- **Oligomeric state:** BC functions as a **homodimer**. UniProt annotates the assembled ACC as "a **heterohexamer of biotin carboxyl carrier protein, biotin carboxylase and the two subunits of carboxyl transferase in a 2:2 complex**." Isolation of the *E. coli* BC·BCCP subcomplex demonstrated that "**the complex consists of a dimer of BC plus four BCCP**

molecules," with the N-terminal ~30 residues of BCCP mediating the BC interaction [PMID 12794081]. Thus BC carries out catalysis as a dimer, tightly coupled to the mobile biotinylated BCCP that ferries the carboxyl group to carboxyltransferase.

5b. Genomic Context in *P. putida* KT2440

In the KT2440 chromosome, *accC* (PP_0558) is **immediately adjacent to *accB* (PP_0559)**, which encodes the **biotin carboxyl carrier protein (BCCP, Q88QD5, 153 aa)** — the very substrate that BC carboxylates. This constitutes the canonical bacterial **accBC gene cluster**, physically coupling the carboxylase (AccC) to its carrier (AccB) (locus mapping from UniProt/KEGG; neighbors PP_0557 *acoR* and PP_0560 *aroQ1*). This genetic linkage mirrors the tight functional coupling of the two proteins and is a hallmark of the **multi-subunit (dissociable) class of ACC (MS-ACC)** used by bacteria and plant chloroplasts, as opposed to the single-chain eukaryotic ACC [PMID 12121720, 34132100].

The biotin prosthetic group that BC acts upon is not synthesized on AccB directly; it is **post-translationally attached to a single, highly conserved lysine of BCCP within an Ala/Val-Met-Lys-Met consensus** by **biotin protein ligase BirA** (EC 6.3.4.15) in an ATP-dependent two-step reaction (biotinyl-5'-AMP formation, then transfer) [PMID 20443544, 15033356]. Thus AccC's physiological substrate is *holo*-BCCP (biotinylated AccB).

6. Pathway Context and Biological Process

- **Pathway:** *De novo* **fatty acid biosynthesis** (type II FAS). BC/ACC produces **malonyl-CoA**, the universal two-carbon donor that is loaded onto acyl carrier protein (via FabD) and iteratively condensed (FabB/F, FabG, FabA/Z, FabI) to elongate acyl chains for **membrane phospholipids**. This is the **committed, rate-limiting entry step** of the pathway [PMID 16707089, 28469974].
- **Cofactor/prosthetic group dependency:** The reaction depends on the **biotin** prosthetic group carried by BCCP (attached by biotin protein ligase, BirA), and on **Mg²⁺** and **ATP**. BC itself does not carry biotin; it modifies the biotin on its partner subunit.
- **Physiological importance:** ACC activity governs the flux of acetyl-CoA into lipids; malonyl-CoA is essential and its synthesis is required for growth. In diverse bacteria ACC

components are **essential genes**, making BC an antibacterial target [PMID 16707089, 28469974].

- **Regulatory note (post-translational):** In *E. coli*, BC (AccC) activity and folding can be modulated by the peptidyl-prolyl isomerase PpiB, which directly interacts with and increases AccC enzyme activity [PMID 27306110] — illustrating that BC function can be tuned at the protein level, though this is a peripheral observation rather than the core catalytic role.

6b. Pseudomonas-Specific Validation and Downstream Metabolic Fate

Beyond the strong cross-species inference from *E. coli*/*H. influenzae*, the AccC = biotin-carboxylase assignment is **experimentally supported within the *Pseudomonas* genus**. In a FAS II reporter screen in *P. aeruginosa*, genetic and biochemical follow-up of sulfonamidobenzamide inhibitors showed they "**target the enzyme (AccC) catalyzing the biotin carboxylase half-reaction of the acetyl coenzyme A (acetyl-CoA) carboxylase step in the initiation phase of FASII in *E. coli* and *P. aeruginosa***" [PMID 26169404]. Given that *P. putida* AccC is 70.5 % identical to *E. coli* AccC with a fully conserved active site, this genus-level evidence applies directly to PP_0558.

Downstream fate of the malonyl-CoA product. The malonyl-CoA generated by ACC is the committed two-carbon donor for **type II fatty-acid synthesis (FAS II)**, producing the acyl chains of membrane phospholipids. In *P. putida* specifically, de novo FAS II output is additionally channeled by the transacylase **PhaG** ((R)-3-hydroxyacyl-ACP:CoA transacylase) into **medium-chain-length polyhydroxyalkanoate (mcl-PHA)** storage polymers — PhaG being "**the only metabolic link between fatty acid de novo biosynthesis and PHA biosynthesis in this bacterium**" [PMID 11425728; see also

P 36763117

— and the related pathway supplies lipid moieties for **rhamnolipids**. Thus accC sits at the metabolic apex feeding membrane lipid, storage-polymer, and biosurfactant biosynthesis in *P. putida*. (These are downstream consequences; the precise role of AccC itself remains the BC half-reaction.)

Drug-target relevance. Because bacterial ACC/BC is essential and structurally divergent from the eukaryotic single-chain enzyme, it is a **validated, selectively druggable antibacterial target**; structural and inhibitor studies distinguish the BC and CT subsites and demonstrate phylum-selective inhibition [PMID 22183085, 22339356].

7. Supported and Refuted Hypotheses

Supported: - H1 — Q88QD6/accC is a biotin carboxylase catalyzing ATP-dependent carboxylation of biotin (EC 6.3.4.14). **Strongly supported** (UniProt EC/pathway annotation; ATP-grasp + biotin-carboxylation domains; 70.5 % identity to characterized *E. coli* BC; full catalytic-residue conservation). - H2 — It performs the first half-reaction of ACC, committing acetyl-CoA to malonyl-CoA/fatty-acid synthesis. **Supported** (UniProt FUNCTION/PATHWAY; literature on ACC as first committed step). - H3 — Mechanism proceeds via a carboxyphosphate intermediate with Glu296/Arg338-equivalent catalytic residues. **Supported** by conserved-residue mapping plus *E. coli*/*H. influenzae* mechanistic and structural studies. - H4 — BC is a cytoplasmic homodimer within a labile multi-subunit ACC complex. **Supported** (UniProt subunit annotation; BC·BCCP stoichiometry study).

Refuted / ruled out: - The gene symbol is **not** ambiguous; no alternative "accC" gene competes for this designation. Q88QD6 is **not** a eukaryotic-type single-chain ACC (in *P. putida* the components are separate polypeptides). BC does **not** itself carry biotin (that is BCCP/AccB).

8. Limitations and Future Directions

- **Direct experimental characterization of the *P. putida* KT2440 protein is lacking** (UniProt evidence level "Predicted"). All mechanistic and structural conclusions are inferred by orthology from *E. coli* and *H. influenzae* enzymes; the very high identity and complete active-site conservation make this inference robust, but species-specific kinetics (Km for ATP/bicarbonate/biotin, kcat) have not been measured.
- No experimental *P. putida* BC structure exists; residue numbering was mapped by pairwise alignment rather than superposition. An AlphaFold model + structural superposition would further confirm the ATP-grasp geometry.
- **Future work:** heterologous expression and steady-state kinetics of PP_0558; confirmation of essentiality by conditional knockout in KT2440; complex-assembly/stoichiometry assays

with the cognate AccB/AccA/AccD; and evaluation as an antibacterial target given ACC's validated druggability.

References (PMIDs)

- **26169404** — Wallace et al. (2015) FAS II reporter screen; genetic/biochemical proof that AccC is the BC-half-reaction target in *E. coli* and *P. aeruginosa*.
- **11425728** — Rehm, Mitsky & Steinbüchel (2001) PhaG links de novo FAS to mcl-PHA/rhamnolipid synthesis in pseudomonads.
- **36763117** — Liu et al. (2023) β -oxidation-PHA relationship in *P. putida* KT2440 (PhaG/PhaJ routes).
- **22183085** — Polyak et al. (2012) Structure, function and selective inhibition of bacterial ACC (BC vs CT subsites).
- **22339356** — Luo et al. (2012) ACC inhibitors and therapeutic potential (BC/CT targeting).
- **34132100** — Cronan (2021) *E. coli* acetyl-CoA carboxylase: authoritative modern review of subunit structure, stoichiometry, and regulation.
- **12121720** — Cronan & Waldrop (2002) Multi-subunit acetyl-CoA carboxylases; MS-ACC vs single-chain ACC; mechanism/regulation review.
- **20443544** — Healy et al. (2010) BCCP biotinylation; conserved Ala/Val-Met-Lys-Met attachment motif.
- **15033356** — Brown, Cronan et al. (2004) Biotin repressor / two-step BirA biotinylation of BCCP.
- **20158506** — Molina-Henares et al. (2010) Genome-wide conditionally essential genes in *P. putida* KT2440 (metabolic context).
- **11346647** — Sloane et al. (2001) Site-directed mutagenesis of ATP-binding residues of biotin carboxylase; ATP-grasp fold; Lys116/Lys159/His209/Glu276.
- **19213731** — Chou, Yu & Tong (2009) Crystal structure of *E. coli* BC with substrates; Glu296 general base, Arg338 stabilizes enolate biotin.
- **26020841** — Broussard et al. (2015) Structural analysis of substrate/intermediate/product binding in *H. influenzae* BC; carboxyphosphate $\rightarrow$ CO₂ transfer.
- **2976600** — Ogita & Knowles (1988) Carboxyphosphate intermediacy in biotin-dependent carboxylation (¹⁸O tracer).

- **2971391** — Tipton & Cleland (1988) Steady-state kinetics of BC; random substrate addition; catalytic base pK 6.6.
- **12794081** — Choi-Rhee & Cronan (2003) The BC·BCCP complex of *E. coli* ACC; BC dimer + 4 BCCP.
- **16707089** — Santoro et al. (2006) ACC as the first committed step and antibacterial target; component overview.
- **28469974** — Bennett & Högbom (2017) Structure of essential biotin-dependent carboxylase AccA3; ACC = committed step of type II FAS.
- **27306110** — Skagia et al. (2016) PpiB modulates AccC activity/folding (post-translational regulation).

Analysis performed in Iteration 1: UniProt/KEGG annotation retrieval for Q88QD6, literature synthesis, and a Needleman–Wunsch alignment to E. coli AccC (P24182) confirming 70.5 % identity and strict conservation of all catalytic residues.